

Myles Swamy

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EDUCATION

- **University of Sheffield** 2023 - 2026
*BSc Computer Science - Grade: **First-Class Honours***
- **Xaverian College** 2021 - 2023
*Computer Science, Maths, Physics - A-Level Grades: **AAB***
- **Manchester Enterprise Academy** 2016 - 2021
*GCSE Grades: **9,9,9,9,9,8,5,5** including Mathematics (9) and English (5)*

SKILLS

- **Languages:** Java, Python, Ruby, C#, C++, JavaScript/TypeScript, Kotlin, HTML, CSS, LaTeX, Lua, Haskell
- **Frameworks:** Rails, React, Jetpack Compose, NodeJS, JFrame, TailwindCSS, Spring Boot
- **Tools:** Git, SQLite, MySQL, Visual Studio Code (VS Code), IntelliJ IDEA, AI/LLMs
- **Platforms:** Windows, WSL (Linux), Web, Arduino
- **Soft Skills:** Organisation, Teamwork, Responsibility, Active Listening, Communication, Problem Solving

PROJECTS AND EXPERIENCE

- **Prototype Employee Management System for Ziipline** Spring 2025
Ruby on Rails, Web Development
 - Collaborated within a diverse Agile team in university to engineer a prototype Employee Management System for an industry client.
 - Facilitated regular client meetings to gather dynamic requirements, integrate feedback, and ensure the product aligned with their needs.
 - Delivered and demonstrated the final software solution to the clients via an in-person live demonstration.
- **Reengineering an Open-Source Software Project** Spring 2026
Python, GitHub, Code Review
 - Analysed and tested 'Glances', a live open-source GitHub repository, alongside a semi-experienced university team to identify areas for improvements to the codebase.
 - Applied reengineering patterns, static analysis, and dynamic analysis to locate inefficiencies, code smells, and architectural flaws.
 - Refactored the codebase and implemented comprehensive regression testing to guarantee the project maintained core functionality and stability.
- **VS Code Extension that Gamifies Unit Testing** Autumn 2025 - Summer 2026
TypeScript, Python, GitHub API, Software Testing
 - Engineered a full-stack VS Code extension utilising TypeScript for the frontend and a Python API backend to gamify developers' unit testing workflows.
 - Implemented an asynchronous multiplayer game loop where an 'Attacker' writes failing tests to challenge a function, and a 'Defender' refactors the code to pass them.
 - Integrated repository-specific leaderboards that dynamically track and display user experience points (XP) based on their testing and debugging actions.
- **Pseudo-3D Raycasting Engine Using the DDA Algorithm** Summer 2026
Java, JFrame, Graphics, I/O
 - Developed a pseudo-3D raycasting engine from scratch using Java and JFrame to render a 3D environment from a 2D perspective.
 - Implemented the Digital Differential Analyzer (DDA) algorithm to ensure highly efficient wall detection mechanics.
 - Engineered a robust parsing system that generates a navigable 3D world based on a simple, user-editable. plain text file.
- **Real-Time Sensor Dashboard Using Readings From an ESP32 Circuit** Spring 2026
Internet of Things, C++, Arduino
 - Worked in a pair to design and build an IoT hardware system utilising an ESP32 micro-controller, LEDs, and an analogue temperature sensor.
 - Programmed the ESP32 in C++ to transmit live sensor telemetry to a custom web server for real-time graphical monitoring.
 - Developed a two-way communication dashboard allowing users to trigger various LED patterns that dynamically adjust their speed based on ambient temperature readings.

ADDITIONAL INFORMATION

- Rapidly prototyped and delivered a fully playable game within 48 hours during a Game Jam in a 3-person team using Godot.
- Volunteer at a local community grocery.
- Speedcuber that can solve the 3x3 Rubik's Cube in under 20 seconds.
- Enjoy Dungeons & Dragons for its emphasis on collaborative problem solving and team strategy.
- Provided weekly support for a local church, which included live video mixing and performing in the band.